Smart College Complaint Managemen

STUDENT-FRIENDLY GUIDE

SmartCMS - Full Project Summary & Easy Explanation Guide

1. What is this Project? (The Big Idea)

SmartCMS is an easy-to-use college complaint website built for students, teachers, repair workers, and college leaders. In many colleges, when a fan stops working or a water tap breaks, students have to write on paper forms or run around offices. Complaints get forgotten, delayed, or marked 'fixed' even when nothing was done.

SmartCMS fixes all of this. It turns complaint tracking into a clear, fast, and transparent digital process right from your phone or laptop.

Core Purpose in One Sentence

A smart online complaint portal where students report problems, repair staff must prove their work with a photo and a secret code, and college supervisors get alerted if work is not done on time.

2. Who Uses This System? (The 4 Roles)

User Role	Real Person	What They Can Do in SmartCMS
Student	Any college student	Post issues, click 'Me Too', track timer, verify repair with OTP, give star rating
Staff (Worker)	Electrician, Plumber, etc.	See assigned work, start jobs, upload photo proof of fixed item, request OTP
Dept Head / Warden	Professors & Supervisors	Watch over department tickets, receive auto-escalated issues when workers delay
Administrator	Principal / System Admin	Create departments, manage all user accounts, view live charts, inspect audit history.

3. The Most Important Features (Explained in Simple Words)

- **1. 'Me Too' Button (No Duplicate Complaints):** If 5 students have broken fans in the same classroom, they don't need to post 5 separate complaints. They simply click 'Me Too' on the first one! This increases the complaint's urgency score automatically.
- **2. Mandatory Photo Proof:** Repair staff cannot just press 'Done' and walk away. They must take a picture of the repaired item and upload it as real proof before the ticket can move forward.
- **3. 4-Digit Secret OTP Verification:** When the technician finishes, the student gets a private 4-digit code (OTP) on their screen and email. The repair is only closed when the student confirms with their code or leaves a 1 to 5 star review.
- **4. Automatic Timer (SLA):** Every complaint gets an official countdown clock: Emergency (6 hours), High/Medium (24 hours), Low (72 hours). Everyone can see how much time is left.
- **5. Automatic Robot Escalation (Background Worker):** Every 5 minutes, an automatic computer task checks all tickets. If a worker is taking too long or an issue is unassigned, it automatically assigns a free worker and sends an alert to the Department Head!
- **6. Anonymous Option:** Students can check 'Submit Anonymously' for sensitive grievances. Their name and email are kept hidden from public feeds and technician views.
- **7. Live Real-Time Updates:** Thanks to WebSocket technology, status changes, new upvotes, and alerts appear instantly on your screen without having to refresh the browser.

4. Step-by-Step: The Life of a Complaint

Lifecycle Step	What Happens
Step 1: Student Files Complaint	Student enters title, room location, category (e.g., Electrical), and priority. An official Ticket ID like CMP-2026-000001 is created.
Step 2: Auto or Manual Assignment	The system or Department Head assigns the job to an available technician in that department.
Step 3: Work in Progress	The technician marks the ticket 'In Progress' and goes to the room to fix the problem.
Step 4: Photo Proof Uploaded	The technician uploads a photo of the completed repair. The status changes to 'Pending Verification'.
Step 5: Student Verification & Rating	The student checks the repair, enters their 4-digit OTP, and gives a 1 to 5 star rating. The ticket is officially 'Closed'!

5. Technology Used (Explained for Beginners)

- **Frontend Folder (frontend/):** React 18, Vite, and TailwindCSS. Modern, clean dark/light look, smooth buttons, and interactive cards.
- **Backend Folder (backend/):** Node.js and Express 4. Handles user logins, password security, API requests, and email notifications.
- **Database (The Memory):** MongoDB with Mongoose. Saves all students, tickets, proof photos, and activity history.
- **Zero-Friction Fallback:** If MongoDB is not installed locally, the system automatically turns on an in-memory database and loads demo accounts instantly!
- **Instant Messenger:** Socket.io. Sends live alerts and updates directly to user screens like a chat app.
- **Background Clock:** Node-Cron. Runs every 5 minutes in the background to enforce deadlines and auto-escalate delayed issues.

6. Ready-to-Use Demo Accounts

Role	Email Address	Password	Notes
Admin	admin@college.edu	Admin@Password123	Full power, view analytics & audit logs
Chief Warden	chief.warden@college.edu	Warden@Password123	Supervisor for major escalations
Dept Head	elec.head@college.edu	Staff@Password123	Electrical Department Head
Technician	elec.tech@college.edu	Staff@Password123	Electrician maintenance worker
Student 1	alex.student@college.edu	Student@Password123	Standard student account
Student 2	priya.student@college.edu	Student@Password123	Has sample ticket with OTP: 4321

1-Click Login Tip

On the login page (http://localhost:5173/login), you do not need to type passwords manually. Simply click any of the 1-Click Demo Buttons at the bottom to test each role instantly!